KSISU: Evidence-First District Skill Intelligence

A reproducible method for sparse public labour-market evidence, explicit uncertainty, and data-collection prioritisation.

Manufacturing pilot **NCO scope:** 72, 74 **Reference FY:** 2022-23 **Monte Carlo:** 5,000 shared draws

What this prototype demonstrates	What it deliberately does not claim
Real DGT 2026 provisional district x ITI x trade supply enters the evidence pipeline	A proxy ranking is a proven district skill-gap result
Demand and supply are kept on separate evidence ladders	Partial DGT coverage is a complete district census
Uncertainty, evidence gates and human review are explicit	An official Karnataka Skill Index formula
Outputs include value-of-information for future onboarding	A policy recommendation without full district evidence

Core message: the system replaces proxy supply with real public evidence wherever available, but refuses to turn incomplete evidence into a policy call.

1. Evidence ladder and data grain

Input	Grain	Period	Role
LGD district register	district	2026	canonical geography / join key
Udyam district employment	district	2022 / Apr-2023 publication	district allocation proxy; all-sector MSME employment
ASI Karnataka manufacturing	state	2022-23	state manufacturing employment base
NCVT MIS Karnataka ITI magnitude	state	2022	state seat magnitude / conservation benchmark
Historical ITI counts	district weight	2022	historical allocation weights only
DGT CTS-2026 provisional seat matrix	district x ITI x trade	2026	real public district/trade supply evidence
Occupation mix	state prior	2022-23	analyst prior until PLFS occupation-by-industry microdata is loaded
Trade -> NCO crosswalk	trade -> NCO	versioned	human-reviewed pilot mapping

Run	Rows	Seats	Coverage	Policy status
Proxy baseline	60 analytical cells	proxy-calibrated	No district actual supply	60 NEEDS DATA
DGT share-calibrated upgrade	60 analytical cells	45,416 published DGT seats	58 / 60 district-group pairs; 88.0% in pilot scope	60 NEEDS DATA

All outputs carry source provenance, year, grain and evidence status. The DGT 2026 matrix is explicitly provisional in the official source and is not treated as final seat capacity.

2. Inputs in the workbench

The workbench makes every research input visible before any analysis is run. This is the audit surface: a judge can see the source file, grain and period instead of trusting a hidden preprocessing step.

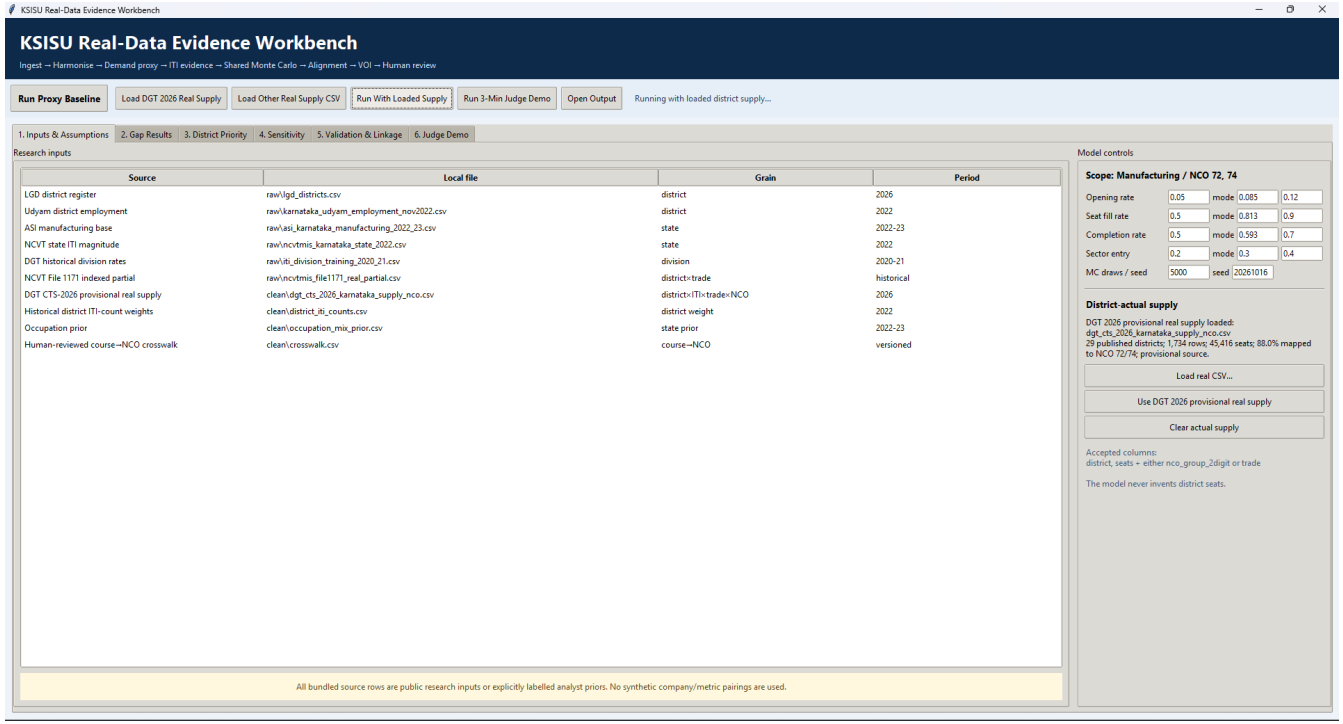


Figure 1. Local research inputs and model controls after the DGT provisional supply source is selected.

Visible control	Current value	Interpretation
Opening rate	5% / 8.5% / 12%	analyst range for annual openings
Seat fill	50% / 81.3% / 90%	training utilization prior
Completion	50% / 59.3% / 70%	completion prior
Sector entry	20% / 30% / 40%	analyst prior pending district placement data
MC draws / seed	5,000 / 20261016	reproducible stochastic run

3. Demand model - allocator choice is a structural sensitivity

District manufacturing employment is allocated from the Karnataka ASI manufacturing total using an explicit Udyam-based district share. The default run uses Udyam employment share; enterprise-unit share and a 50/50 blend are also tested because demand allocation is the weakest structural input.

NCO group	Share of total manufacturing employment	Status
72	0.25	modeled
74	0.15	modeled
Outside pilot	0.60	not modeled

Demand allocator	Definition	Use
udyam_employment	Udyam employment share	default baseline
udyam_units	Udyam enterprise-unit share (June 2026)	structural sensitivity
blend	50/50 average of the two normalized shares	structural sensitivity

Allocator stability diagnostic	Value
Default allocator	employment share (default)
Mean district rank range across allocators	3.70
Districts top-5 across all allocators	2
Within-allocator lognormal sigma	0.35 analyst assumption; no district calibration yet

Demand formula: D(d,o) = ASI manufacturing base × selected district allocator share × occupation share; O(d,o) = D(d,o) × opening rate. The selected NCO shares remain 0.25 + 0.15 = 0.40 and are not normalized.

Interpretation: a district priority is called structurally robust only when it remains high across the employment, enterprise-unit and blended demand allocators. The 0.35 lognormal term captures uncertainty within an allocator; allocator choice itself is tested separately.

4. Real DGT 2026 district x ITI x trade supply - evidence upgrade

The official DGT CTS-2026 provisional dynamic seat matrix is the strongest public supply input in this build. The source states that the tentative matrix was generated from approved instructor mappings as of 08.07.2026 and remained subject to change until the 20.07.2026 cut-off. The package therefore labels it provisional, not final.

DGT evidence item	Result	Meaning
Karnataka rows	1,734	real published ITI-trade rows in uploaded DGT PDF extraction
Distinct ITIs	588	ITI MIS codes represented
Published districts	29	29 in the provisional extraction
Published provisional seats	45,416	actual DGT seat-capacity rows
NCO 72 + 74 mapped seats	39,984 (88.0%)	pilot crosswalk scope
OUT_OF_SCOPE seats	5,432 (12.0%)	kept in total; excluded from the 72/74 pilot
UNMAPPED seats	0 (0.0%)	still requires human review; currently zero in the expanded pilot crosswalk
Analytical pair coverage	58 / 60	Kodagu is missing from the DGT provisional file; Ballari covers the merged Ballari-Vijayanagara geography
Policy gate	NEEDS DATA	coverage is incomplete / provisional

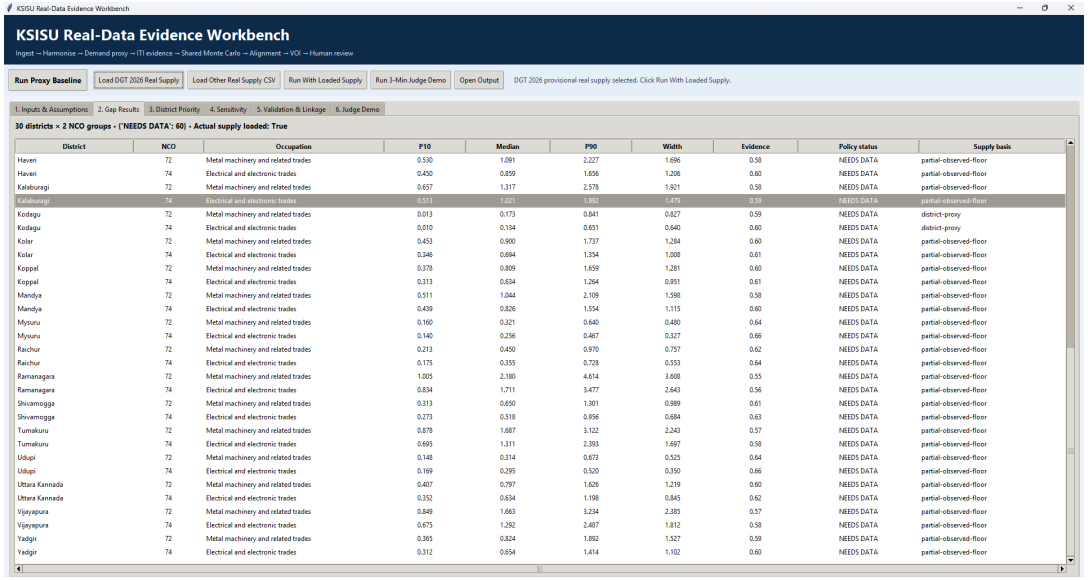


Figure 2. Real DGT supply loaded into the model. Tumakuru and Kalaburagi show district-level provisional evidence, but the policy gate remains closed because coverage is incomplete.

5. Supply model and scale reconciliation

Supply is kept as a flow: seats x fill rate x completion rate x sector entry x trade-to-NCO weight. The DGT matrix is not silently substituted for the NCVT state magnitude; the two sources are reported side by side so the scale difference is visible.

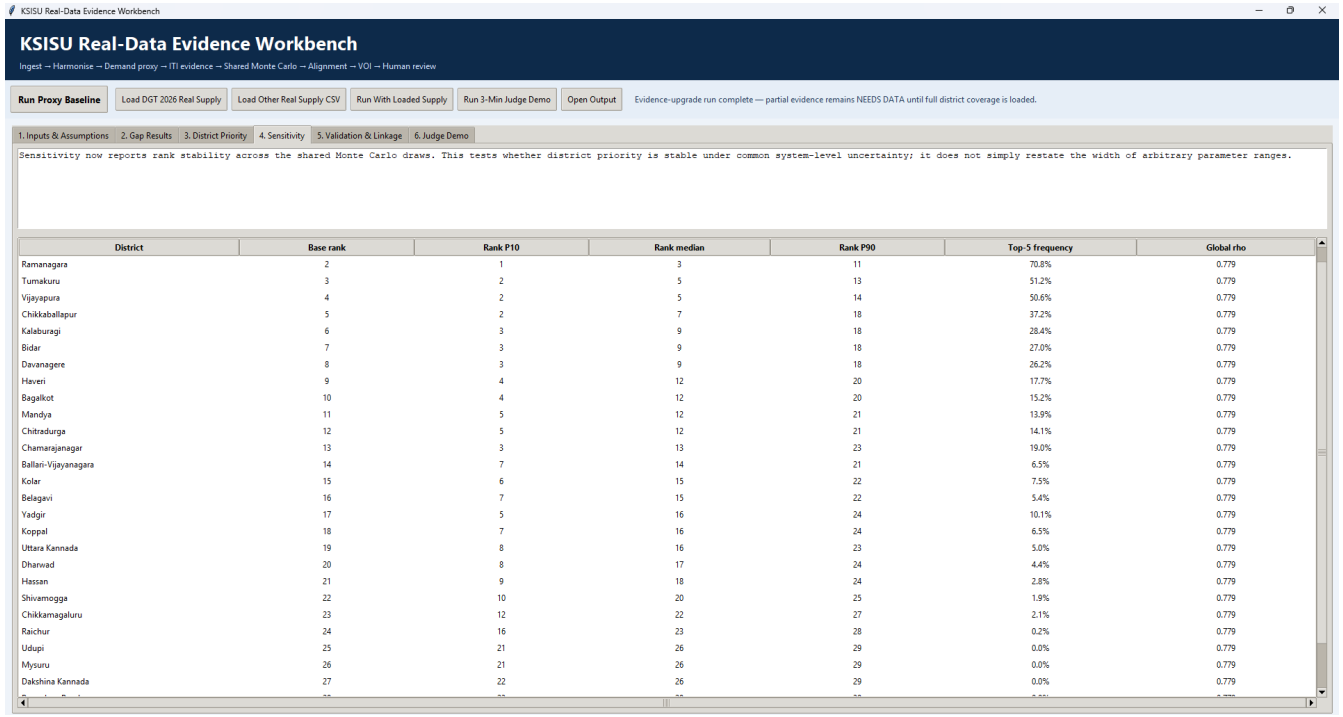
Scale check	Value	Interpretation
NCVT Karnataka state seats (2022)	216,788	statewide magnitude benchmark
DGT CTS-2026 provisional seats	45,416	real published provisional district/trade rows
DGT / NCVT seat ratio	0.209 (20.9%)	WARNING: below configured 0.50 reconciliation threshold
DGT ITI / historical ITI coverage	6e+02 ITIs; mean coverage 40.1%	approximate cross-period coverage weight; not exact census coverage
Mapped to NCO 72/74	39,984 (88.0%)	within pilot scope
OUT_OF_SCOPE	5,432 (12.0%)	not a mapping failure; outside current pilot
UNMAPPED	0 (0.0%)	true mapping/review remainder

Why the scale warning matters: 45,416 published DGT seats represent only about one-fifth of the 2022 Karnataka NCVT seat magnitude. The DGT rows are therefore used primarily as a **distribution signal** across districts and NCO groups, not as the statewide level. District DGT shares are shrunk toward the historical ITI-count proxy in proportion to approximate matrix coverage (DGT ITIs / 2022 ITIs); a district absent from the matrix therefore retains its proxy rather than receiving zero. The allocation concentration is intentionally lower for this provisional evidence mode.

Coverage rule	Current state
Required analytical district x NCO pairs	60
Covered by DGT provisional file	58
Full-evidence gate	CLOSED
Policy decision status	60 NEEDS DATA

6. Relative alignment and data-collection priority

SCI and VOI are presented as prioritisation tools. They are not validation of the district ranking because missing district/group evidence still falls back to proxy allocation. The ranking is based on the uncapped median ratio R; SCI is shown separately as a capped coverage index.



District	Base rank	Rank P10	Rank median	Rank P90	Top-5 frequency	Global rho
Ramanagara	2	1	3	11	70.8%	0.779
Tumakuru	3	2	5	13	51.2%	0.779
Vijayapura	4	2	5	14	50.6%	0.779
Chikkaballapur	5	2	7	18	37.2%	0.779
Kalaburagi	6	3	9	18	28.4%	0.779
Bidar	7	3	9	18	27.0%	0.779
Davanagere	8	3	9	18	26.2%	0.779
Haveri	9	4	12	20	17.7%	0.779
Bagalkot	10	4	12	20	15.2%	0.779
Mandya	11	5	12	21	13.9%	0.779
Chitradurga	12	5	12	21	14.1%	0.779
Chamarajanagar	13	3	13	23	19.0%	0.779
Ballari-Vijayanagara	14	7	14	21	6.5%	0.779
Kolar	15	6	15	22	7.5%	0.779
Belagavi	16	7	15	22	5.4%	0.779
Yadgir	17	5	16	24	10.1%	0.779
Koppal	18	7	16	24	6.5%	0.779
Uttara Kannada	19	8	16	23	5.0%	0.779
Dharwad	20	8	17	24	4.4%	0.779
Hassan	21	9	18	24	2.8%	0.779
Shivamogga	22	10	20	25	1.9%	0.779
Chikkamagaluru	23	12	22	27	2.1%	0.779
Raichur	24	16	23	28	0.2%	0.779
Udupi	25	21	26	29	0.0%	0.779
Mysuru	26	21	26	29	0.0%	0.779
Dakshina Kannada	27	22	26	29	0.0%	0.779

Figure 3. District priority view. VOI answers where better district evidence should be collected first; it is not a claim that the top-ranked district has the largest true skill shortage.

Interpretation rule	Use
Median R	alignment signal; can be above or below 1
SCI	share of modeled openings covered at min(R,1)
VOI	interval width x annual openings
Data priority	VOI adjusted by evidence/confidence; onboarding queue

7. Sensitivity as rank stability

Shared Monte Carlo draws are used across districts. Rank stability is a common-system uncertainty diagnostic. The stronger judge-facing test is the six-scenario robustness table, which compares proxy and DGT-share supply routes under all three demand allocators.

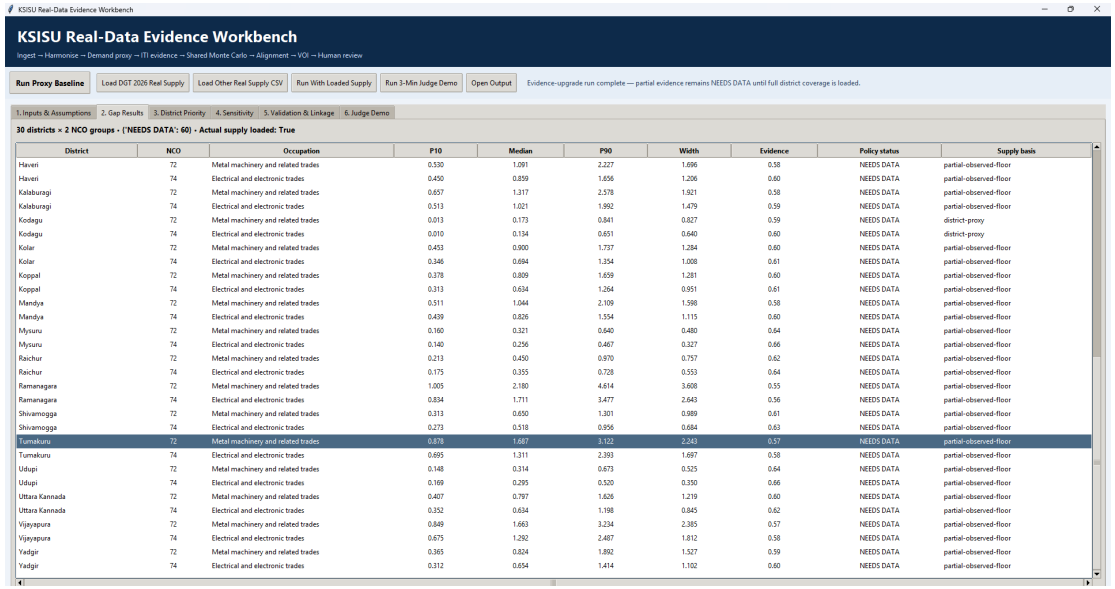


Figure 4. Rank stability under shared uncertainty. The same draw structure is used across districts so relative ranking is comparable.

District	Top-5 appearances / 6	Rank min-max	Robust across all six
Vijayapura	6	2-4	YES
Gadag	5	1-6	No
Tumakuru	5	1-6	No
Ramanagara	4	2-8	No
Bagalkot	3	1-10	No

Six-scenario result: Vijayapura is top-five in all six scenarios; 16 districts have a rank range above 5. A robust flag is therefore a stress-test result, not a declaration of a true skill shortage. The 50/50 blend is an arbitrary midpoint sensitivity case, not an empirical estimator. The 0.35 lognormal demand term is an explicit assumption with no district-level calibration yet.

8. Component B - refresh automation, lineage and human review

Component B turns scheduled refresh from a narrative claim into an executable gate. Each registered source file is hashed, its schema and row count are checked, statewide NCVT seat movement is tested against a 20% anomaly threshold, and any review item stops publication until a human approves the update.

Automation check	Current run	Judge meaning
Registered files checked	18	source registry is machine-audited
Unchanged / skipped	18	no unnecessary reprocessing
Changed files in current baseline	0	none in the frozen snapshot
Statewide seat anomaly threshold	20%	a large source shift requires review
Empty review queue output	header preserved	machine-readable even when no rows need review
Legitimate source update	`python refresh_pipeline.py --make-baseline`	approved update resets the comparison baseline

Evidence claim	Verification
Historical ITI allocation weights	30 / 30 analytical rows match Rajya Sabha Annexure-IV page 6; printed 275 Government + 1,227 Private = 1,502 total
Ballari-Vijayanagara public export	2,268 observed NCO 72/74 DGT seats; ratio and rank remain visible
Kodagu public export	No DGT rows; explicitly `not_observed` and retains modeled historical-ITI-weight supply
Suppression guardrail	Institutional aggregate data only; no personal data is ingested. Threshold 5 is a safeguard for future finer-grained data

The same refresh layer also records SHA-256 provenance, source URL, period and grain. Human review is a required publication step, not an exception path.

9. Judge demo: what KSISU can see, and what it should collect next

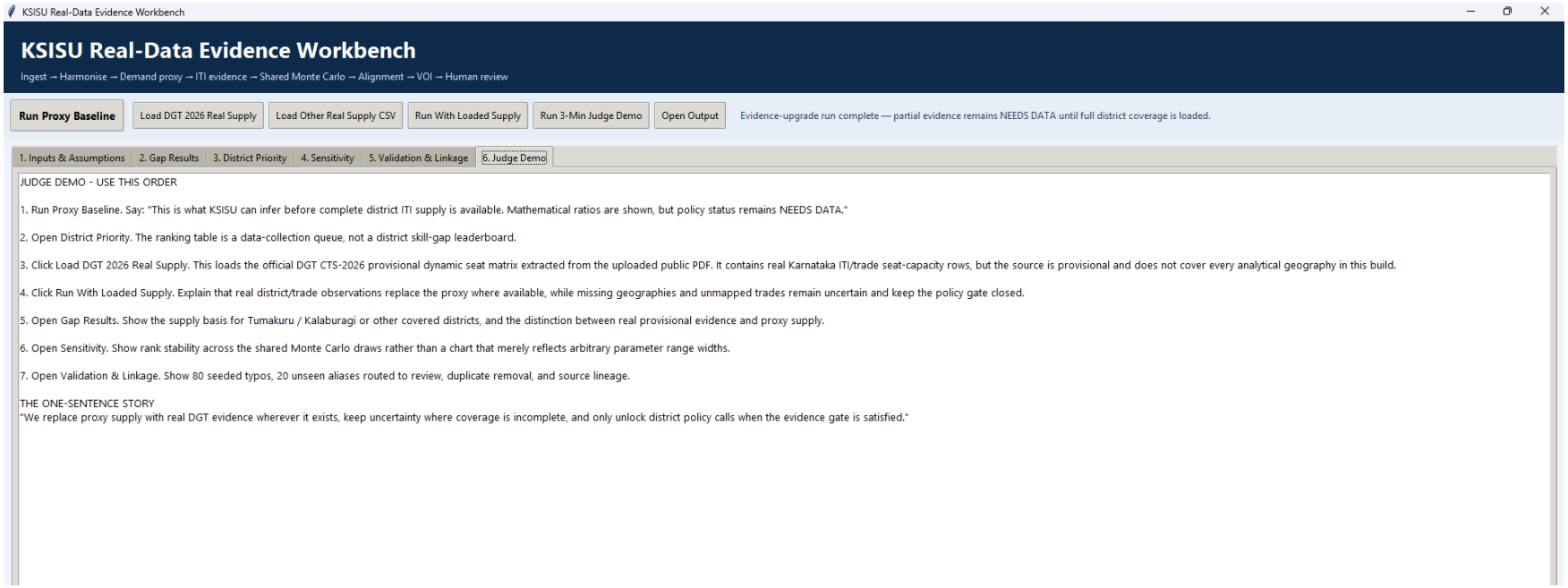


Figure 6. Judge-demo sequence embedded in the workbench.

Demo step	What to say
1. Proxy baseline	This is the conservative public-data baseline.
2. District priority	This is a data-collection queue, not a district skill-gap leaderboard.
3. Load DGT 2026 real supply	Real district x ITI x trade rows replace proxy supply where they exist.
4. Run with loaded supply	Partial coverage remains an observed floor plus uncertain remainder.
5. Gap results	Show P10 / median / P90 and the policy gate.
6. Sensitivity	Show whether priority is stable across shared draws.
7. Validation	Show typo recovery, review queue, dedupe and provenance.

One-sentence story: “We use real DGT district/group seat distribution to improve the allocation, keep uncertainty where coverage is incomplete, and only unlock a district policy call when the evidence gate is satisfied.”

Current blocker: the public DGT 2026 matrix is provisional, covers 58/60 analytical district-group pairs, and its 45,416 published seats are only 20.9% of the 2022 NCVT state seat magnitude. The DGT-share run changes the district allocation using coverage-weighted DGT shares (588 DGT ITIs relative to the historical 2022 denominator), while the demand allocator is stress-tested across Udyam employment, Udyam units and their blend. The model still makes no district policy claim because evidence coverage is incomplete.

Submission note: SCI, VOI, confidence and evidence gates are proposed methodology for the Skillathon, not an official Karnataka formula. The DGT source is a provisional 2026 dynamic seat matrix, not a final frozen capacity table.